



Paddock design influences soil inorganic nitrogen distribution in a pasture-based sow system with poplar trees

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Abstract In pasture-based pig production, high soil inorganic nitrogen (N-inorg) amount and uneven distribution combined with low vegetation cover lead to nitrate leaching. Integrating trees can mitigate this by maintaining a vegetation ensuring nitrogen (N) uptake, but there is limited knowledge on how to implement this approach. An on-farm experiment was conducted using 16 single sow paddocks with grass-clover ley and centrally located poplar trees to investigate the effect of paddock design on elimination behaviour, area covered with faeces and herbaceous vegetation, and distribution of N-inorg. Four treatments were evaluated in three consecutive sow batches, involving 48 lactating sows, combining 1) tree management—cutting trees with application of wood chips as mulch under the trees or unmanaged trees; and 2) sow resource position—placing the

hut and feeding trough on the same side or opposite sides of the trees. Topsoil N-inorg concentrations were sampled initially, after two, and after three sow batches. Elimination behaviour and area covered with faeces indicated higher N deposition within the tree zone compared to the grass zones, with no clear treatment effect. However, N-inorg was more evenly distributed in paddocks with resources on opposite sides of the trees by ranging at 41.9–44.2 kg N ha⁻¹ instead of 23.8–76.1 kg N ha⁻¹. Higher N-inorg amount was associated with lower herbaceous vegetation cover. Tree management had no effect. These findings suggest that a paddock system with centrally placed poplar trees and huts/feeders on either side may enhance the distribution of soil N-inorg and potentially reduce N leaching.

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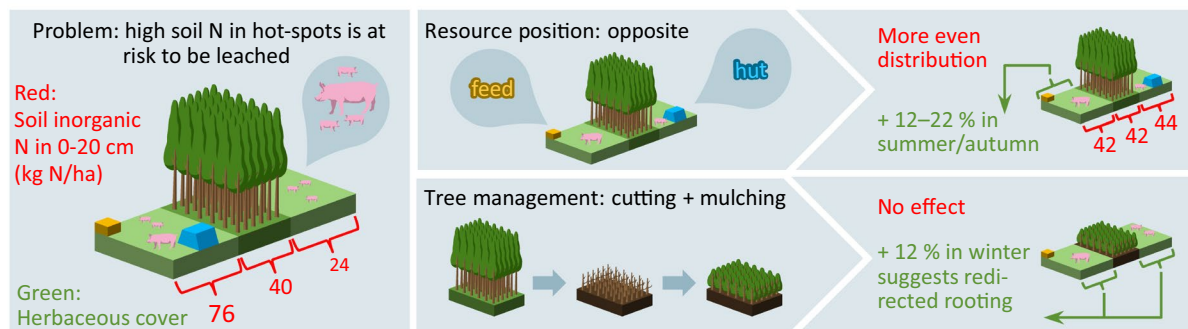
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Graphic Abstract

Soil inorganic nitrogen distribution in a pasture-based sow system with poplar trees

Resource position can reduce soil inorganic N hot-spots



Keywords Soil mineral N · Pig production · Silvopasture · Agroforestry · Elimination behaviour · Paddock management

Introduction

Animal production is associated with high nitrogen (N) surpluses (Jan et al. 2017) and thus the environmental challenge of N losses. These challenges are further complicated by free-range production systems implemented to provide animal benefits. Surface N balances for free-range pig and poultry paddocks can easily exceed 300 kg N ha⁻¹ in a year (Kongsted et al. 2019; Manevski et al. 2024; Williams et al. 2000). This N surplus is mostly added by the animal manure and is subject to gaseous N emissions (ammonia (NH₃) volatilisation, nitrous oxide (N₂O) from denitrification) or nitrate (NO₃⁻) leaching, and only a small part is integrated in the soil organic matter pool. Under temperate wet climate and on sandy soils, such as in Denmark and parts of England, up to 40–90% of the N surplus is leached from free-range pig paddocks (Eriksen 2001; Williams et al. 2000). Of this, more than 200 kg N ha⁻¹ can leach during one year with sows (Eriksen 2001; Kongsted et al. 2020). Leached NO₃⁻ is hazardous to groundwater quality (Schullehner et al. 2018) and contributes to the eutrophication and oxygen deficiency of surface waters (Davie and Quinn 2019; Meier et al. 2019).

Environmentally sustainable free-range pig production therefore remains challenged, and sound measures are needed to reduce the high surplus of N.

This challenge is exacerbated by the pigs' spatially concentrated elimination behaviour (Eriksen 2001; Salomon et al. 2007), leading to N 'hotspots' of about 1,800 kg N ha⁻¹, at a short distance around the hut and the feeding area (Salomon et al. 2007). Herbaceous vegetation cover acts as an N sink, but is typically destroyed by rooting and trampling, so that large quantities of inorganic N remain in the soil (Eriksen et al. 2002). Equipping sows with nose-rings is aimed at maintaining the grass cover but cannot fully achieve it (Eriksen et al. 2006). Woody perennials, on the contrary, are more robust and can withstand pig damage once well established (Horsted et al. 2012; Jørgensen et al. 2018). In addition to adsorbing some N which would otherwise leach (Manevski et al. 2019), trees benefit the pigs by providing shade and a more natural environment (Jakobsen 2018; Schild et al. 2018). By redirecting the pigs' activity, trees might help maintain the grass cover and thus interactively contribute to soil N uptake (Horsted et al. 2012; Jørgensen et al. 2018). Trees also take up N during the non-growing season (Ma et al. 2021) with their wider and deeper root system compared to herbaceous vegetation (Fortier et al. 2013; Manevski et al. 2019). When integrating trees, European farmers strive to continue receiving subsidies under the Common Agricultural Policy of the EU (CAP). Energy tree crops must be cut down after a specific time (Regulation 2022/C 485/01,

2022, Point 2.4), e.g. after at most 20 years for poplars in Denmark (Landbrugsstyrelsen 2022). The regrowing shoots provide a sheltered environment for the pigs and are expected to increase N demand of the poplar trees. The harvested material, if not used for energy, can be chopped and used as wood chip mulch (Barrett and Burke 2000; Tang et al. 2024). Such an organic material rich in carbon (C) can increase soil C/N ratio and thus reduce soil inorganic N by providing C for microbial immobilisation of N (Seneviratne et al. 1997; Sønsteby et al. 2004). However, the effect of trees and wood chip mulch on soil N distribution in free-range animal paddocks remains poorly studied.

The magnitude and the evenness of N distribution in free-range animal paddocks depend on stocking density and feed input (Eriksen et al. 2006; Quintern and Sundrum, 2006). A high stocking density and high N excretion from inefficient feeding lead to unevenly distributed N with 'hotspots' that are difficult to control. Directing the pigs' eliminations to the tree area might enhance N distribution and improve its uptake by the trees. However, current knowledge on trees' effects on pig elimination behaviour and N cycling within pasture paddocks is limited to a few studies in pigs (Horsted et al. 2012; Jakobsen et al. 2018; Jørgensen et al. 2018; Manevski et al. 2019) and comparable work in poultry (Manevski et al. 2024) or cattle (Dahlgren et al. 1997). In Denmark, an experimental study indicated that poplar trees reduce NO_3^- leaching by 75–80% in the tree area compared to the grass-clover area (Manevski et al. 2019). In the same experiment, the sows did not strongly prefer to eliminate in the tree area compared to the pasture (Jakobsen et al. 2018). Pigs are known to place their eliminations in relation to the location of resources like the hut and the feeding trough, understood here as resources from the pigs' perspective (Andersen et al., 2020; Salomon et al. 2007; Stolba and Wood-Gush 1989). This explains why, in another experiment, growing-finishing pigs preferred the tree area located between key resources (Horsted et al. 2012). Hence, the paddock design in terms of resource position is relevant for the distribution of pig excreta and thus soil N distribution and management (Jakobsen et al. 2018).

The objective of this study with free-range lactating sows in paddocks with grass-clover and poplar trees was to evaluate the effects of novel management practice on soil inorganic N and its spatial

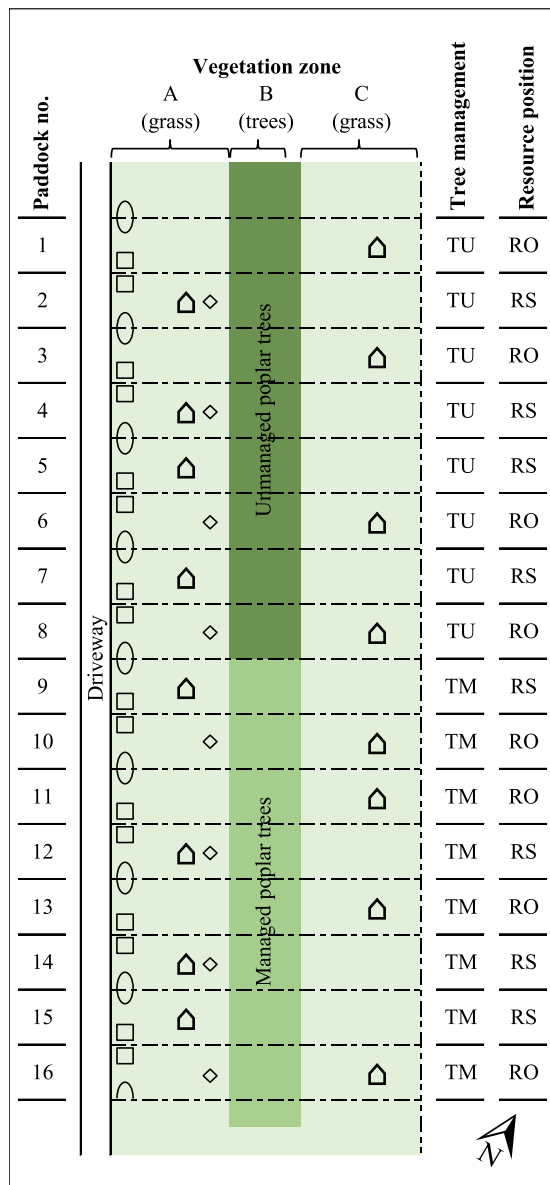
distribution. The treatments included the tree management referring to using the harvested tree biomass as wood chip mulch, and different resource positions. The connection between these treatments and soil inorganic N was investigated by quantifying the elimination behaviour, area covered with faeces, and herbaceous cover. Higher levels of soil inorganic N were expected in areas where animals eliminated, faeces covered the soil and herbaceous vegetation cover was sparse. It was hypothesized that placing the resources hut and feeding trough on opposite sides of the tree zone will distribute the inorganic N within the paddock compared to placing the resources on the same side of trees. Additionally, harvesting and adding wood chips to the soil in the tree area was hypothesized to reduce the inorganic N amounts in this vegetation zone compared to no tree management.

Materials and methods

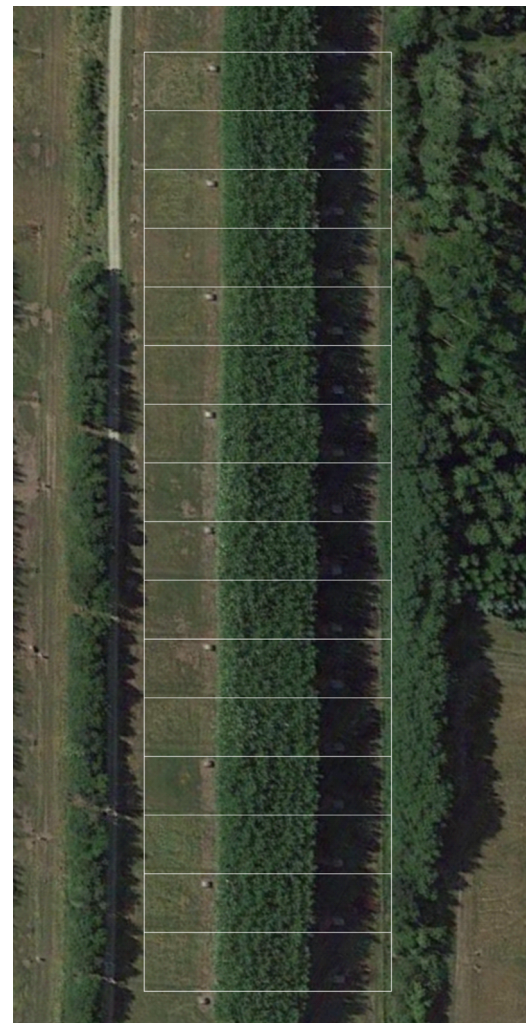
Experimental setup and treatment description

The field experiment was conducted from April 2023 to March 2024 on a commercial organic pig farm in Denmark (Fig. 1). The field consisted of sixteen 15×63 m paddocks, each housing one lactating sow, and with poplar trees centrally placed. This results in a stocking density of 10.6 sows ha^{-1} . These paddocks were used to test four treatments in four replicates. The treatments comprised four combinations of the two factors: 1) tree management (T) where trees were either managed with cutting and placing the wood chips (WC) on the soil surface in the preceding year (TM) or trees were unmanaged (TU) and 2) resource position (R) where hut and feeding trough were either placed on the same side (RS) or opposite sides (RO) of the trees. The paddock area was split in two halves where the tree management treatment was applied (paddock no. 9–16, TM, Fig. 1). The resource position treatments were randomized within half-blocks of the same tree management (Fig. 1).

The poplar clone OP42 (*Populus maximowiczii* × *P. trichocarpa*) was planted in 2011 and occupied about one third of the paddock area. The trees were planted in eight rows crossing the paddocks, with 2 m between rows and 3 m distance between the two middle rows. Within rows, the distance between trees was 2.5 m, i.e. a plant density of about $1,900$ trees ha^{-1} . All the



(a)



(b)

Fig. 1 Setup of the experiment as schematic drawing **a** with 16 paddocks each for one lactating sow comprising four replicates of the four combinations of the two treatment factors tree management and resource position: trees managed (TM) or unmanaged (TU) and hut and feeding station on the same side (RS) or on opposite (RO) sides of poplar trees with indica-

tion of paddock no., vegetation zones and position of huts ($\triangle$), feeding stations (sow: $\square$; piglets: $\diamond$) and water stations ($\circ$). Size not to scale. Satellite photo **b** of the experimental site with indication of approximated paddock borders. Retrieved from Google Maps (2024)

trees in paddocks 9 to 16 were cut down to a height of about 2 m in April 2022 (TM treatment, Fig. 1). The harvested and chopped material was placed on the soil under trees to provide C for biological immobilisation

of inorganic N (Fig. 2). A representative sample of WC contained 16% bark and 84% wood by weight. Particles of the sorting between 1 and 5 cm accounted for 84% of the weight. Details on the WC are available

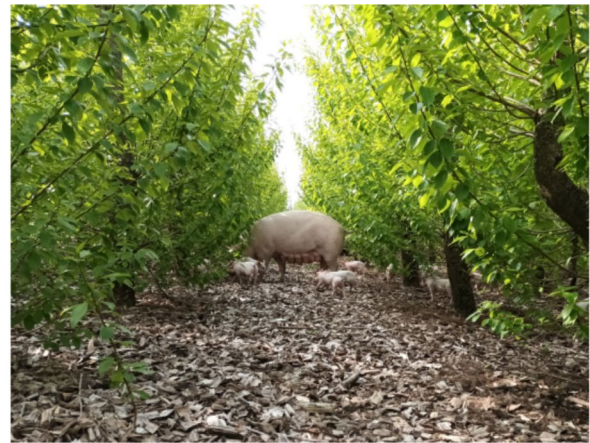


Fig. 2 The environment in the tree zone is very open under unmanaged, tall trees **a**, whereas the new shoots of cut trees create a denser canopy, under which wood chips were placed on the soil surface as the management **b**. The photos were taken in May 2023

in Appendix 2 (Table 3). From measuring the weight of WC harvested from a row of 6 trees and assuming a dry matter (DM) content of 44% (Pecenka et al. 2020) and C and N contents of 43.0% and 1.3% of DM weight (Hadrović et al. 2022), respectively, it was estimated that the applied 105.6 t ha^{-1} WC added 20.0 t C ha^{-1} and 0.6 t N ha^{-1} to the soil in managed tree zones. This results in a C/N ratio of 32.9. Measured from representative trees, the cut poplar regrew on average additional 2 m of height per year during the growing seasons after harvest (2022–2024), whereas the unmanaged trees did not grow substantially. The stem diameter at breast height grew by less than 1 cm in all trees.

During the experiment in 2023, three consecutive batches of 16 sows were each introduced about one week before farrowing in mid-April, end of July and beginning of November, and removed after 10 weeks of lactation (Fig. 3). One sow in batch 1, one in batch 2 and three in batch 3 were removed from the paddocks by the farmers before the end of lactation. As such, a total of 43 sows stayed in the paddocks during the whole lactation period. The sows were of the crossbreed Nordic Landrace x Yorkshire (TN70 by Topigs Norsvin) and in their fourth parity on average, with a range from first to seventh. They were equipped with nose-rings. One-strand electrified wire fenced the sows but not the piglets who could roam freely between paddocks. Each paddock contained a $2.60 \times 2.25 \text{ m}$ farrowing hut, a sow feeding station, and a shared water station (Fig. 1, Fig. 4). Piglet feeding

stations were in every second paddock. Huts were straw-bedded and reallocated after each batch (Fig. 4). On warm days, wallowing water was provided in the grass zone west of the trees (zone C, Fig. 1).

Historically, the farm followed a crop rotation of sows every second year, alternating with spring barley (*Hordeum vulgare*) undersown with grass-clover. In the rotation preceding the measurements, four sow batches were managed: in the first two, huts and feeding troughs were placed on opposite sides of the tree rows; in the last two, both were positioned on the same side.

Data collection

Farm data

To calculate the N mass balance, collected farm data on inputs and outputs included the litter-wise count of born, eliminated and weaned piglets as well as the live weight of weaned piglets (weighed batch-wise) and eliminated piglets (estimated) within the paddocks. On average, the experimental sows weaned 13.0 piglets per litter, with an average individual piglet weight of 29.7 kg. Straw amount and feed N (crude protein) content were retained from farm records. The sow and piglet feed amount was assumed to be the same for each litter according to the farm's feeding curves. Mean sow live weight losses were estimated based on literature (Kongsted et al. 2019).

Weather data

Data on daily mean temperature (°C) and daily precipitation (mm) based on a 20×20 km and 10×10 km grid, respectively, for the years 2014–2024 was attained from the Danish national grid database provided by the Danish Meteorological Institute (DMI).

Elimination behaviour and area covered with faeces

To register sow elimination events, each sow batch was observed during lactation weeks 4 and 7 (Fig. 3). For batches 1 and 2, observations were conducted in two blocks: from 2.30 pm to 7.00 pm on the first day and from 7.00 am to 12.10 pm on the following day with half an hour break per block. For batch 3, both blocks were performed on a single day, and the observation time window was reduced from 7.40 am to 4.30 pm due to morning and evening darkness. Each of the 16 paddocks were observed for similar durations on each observation day. During observations, two observers on opposite sides of the trees observed half of each paddock. They moved along paddocks in consecutive order but with randomisation of starting point and walking direction. The observers systematically switched sides after observing all paddocks once. To provide the time to walk from paddock 1–16 and reverse, a break of 10 min was included. By continuous sampling, urination and defecation events were registered at paddock level, including the sow's position according to the three vegetation zones (Fig. 4). In both week 4 and 7 of the first sow batch, observers spent one sampling interval of 2 min at each sow and paddock before moving to the adjacent one. For the following batches 2 and 3, the interval was prolonged to 4 min to disturb the sows less with two paddocks observed per interval. Leafless trees during batch 3 observations allowed one person to perform the observations alone. In total, each paddock was observed for 3.7 h, resulting from 52, 104, and 108 min per sow in batches 1, 2, and 3, respectively. Only one sow in batch 3 was removed before observations. At the end of each batch, the percentage of area covered with faeces was visually estimated in 1×1 m grids in paddocks 5–12. The faeces of sows and piglets could not be distinguished.

Herbaceous cover

From April 2023 and monthly until the final sow batch, herbaceous cover (%), including grass, clover

and weed, was visually estimated in all 16 paddocks across three subzones within each of the grass zones A and C (Fig. 4).

Soil inorganic nitrogen and texture

Within each of the 16 paddocks, soil samples (at least 200 g) were taken with augers from 0–20 cm depth in vegetation zones A, B, and C using 18, 15, and 18 sampling points, respectively (Fig. 4). Soil inorganic N was determined from samples taken at each point in March 2023 (before sow introduction; date 1), October 2023 (after batch 2; date 2) and March 2024 (after sow batch 3; date 3). Samples were pooled by vegetation zone per paddock resulting in 48 samples per date. These were cooled to −18 °C and dried. Concentrations of NO₃[−] and ammonium (NH₄⁺) in DM were determined spectrophotometrically after extraction with 2 M KCl and summed to represent total inorganic N (N-inorg). Concentrations (mg kg DM^{−1}) were converted to kg N ha^{−1} assuming a bulk density of 1.4 g cm^{−3} for Danish soils presented by Rubaek and Sørensen (2011). For soil texture determination, one sample was taken at each point in March 2023 (Fig. 3), then pooled by vegetation zone and treatment, resulting in 12 samples in total. Soil particle size was determined by soil sieving and hydrometer measurements after dispersion in 0.002 M sodium pyrophosphate (Gee and Bauder, 1986). Total N and C concentrations were determined after air drying at 60 °C and sieving (> 2 mm) by the DUMAS dry combustion using a Vario Max CN Element Analyzer.

Nitrogen mass balance

The excreted N (*ex animal*) was calculated as the difference between feed input and sow weight loss on the one hand and piglet weight gain on the other hand. By including straw for bedding, atmospheric deposition (Ellermann et al. 2024) and biological N fixation (Jakobsen et al. 2015) as further inputs, the surface N balance was calculated for the year with sows (April 2023 to March 2024).

Statistical analysis of sampling date, treatment and zone effects

All statistical analyses were conducted in R statistical software (v4.4.1, R Core Team 2024). To explore

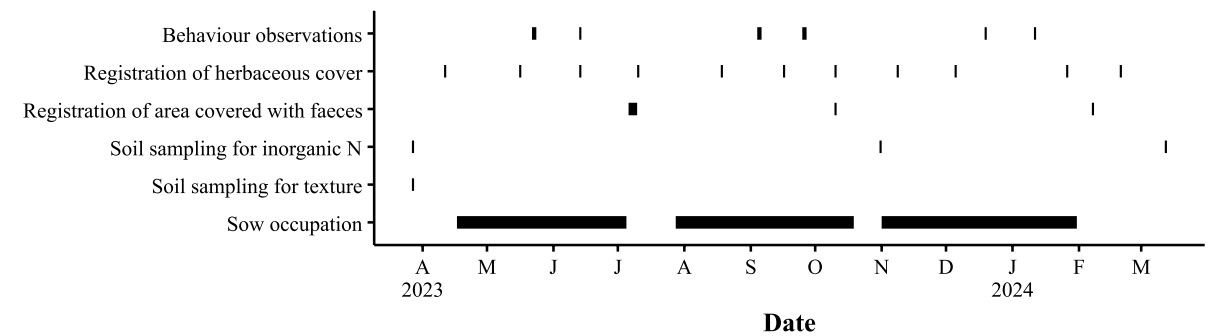


Fig. 3 Timeline of sows occupying the paddocks in three consecutive batches and data collection

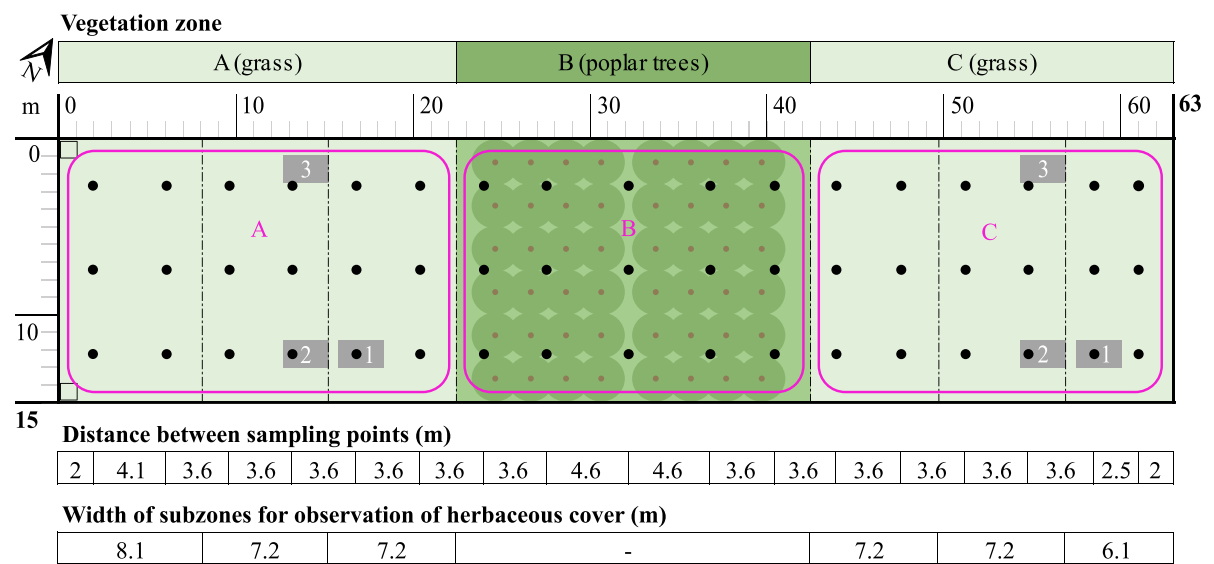


Fig. 4 Schematic drawing of one exemplary paddock with the position of soil sampling points (black dots) and huts (grey boxes). Soil samples were pooled for vegetation zones **A**, **B**, **C** as indicated by the pink encirclements. Huts were located in zone **A** (left) for resources located on the same side of the trees

(**RS**) and in zone **C** (right) for resources on opposite sides (**RO**) and were moved for each batch (no. 1, 2 and 3). The exact location of water and feeding stations (○ in zone **A**) is to be retained from the experiment setup (Fig. 1)

the effects of sampling dates, vegetation zones (“zones”) and treatments on elimination behaviour ($n=141$), area covered with faeces ($n=22,680$), herbaceous cover (1,056), and soil inorganic N ($n=96$ for each NO_3^- , NH_4^+ , and N-inorg), generalized linear mixed effects models (package `glmmTMB`, v1.1.9, Brooks et al. 2017) were fitted. Model specifications were chosen to align with the goal of the analysis and to fit the distribution and the structure of the data best without risking overfitting.

To achieve this, model fittings were evaluated by simulated residual plots from the *DHARMa* package (v0.4.7, Hartig 2022) and fitted values were visually compared to the dataset. As fixed main effects, the zone, tree management, resource position, and batch or date were defined. Interactions of two, three and four of these variables were included if possible. Only for the herbaceous cover, model convergence did not allow to include three- and four-way interactions. For elimination

behaviour, the differences between batches were not in focus and thus the variable was only included in the random effects. The random effects generally account for the specific data structure. To model herbaceous cover, the date was included as a third-degree polynomial term to capture potential non-linear temporal trends across groups. For elimination behaviour counts, a negative binomial probability distribution with a log-link function accounted for overdispersion. The continuous, skewed distribution of faeces and herbaceous cover data, with a high proportion of zeros, was handled using a zero-inflated gamma distribution. In contrast to soil $\text{NH}_4^+\text{-N}$ and N-inorg, the low concentrations of $\text{NO}_3^-\text{-N}$ necessitated a log-link function to meet the assumptions of normality and a dispersal function to model date-dependent heteroscedasticity. Full model specifications are provided in Appendix 1 (Table 2). Model outputs as concentrations of soil inorganic N (mg l^{-1}) were converted to amounts (kg ha^{-1}) as described (Sect. “[Soil inorganic nitrogen and texture](#)”). For each model, the variance explained by each variable was estimated from deviance values using a Wald Chi-square test with type II sums of squares. Only for the area covered with faeces, type III sums of squares were used due to the strong influence of higher-order interactions. Least-square means (LS means) were used to compare the levels of significant interactions of fixed effects.

Correlation analysis between area covered with faeces, herbaceous cover and soil inorganic N

In order to aggregate the repeated measures of area covered with faeces and herbaceous cover at paddock level, paddock was included as a main effect in the generalized linear mixed models described above. The LS-means estimated at paddock-level were used to investigate the association with the soil inorganic N measured at paddock level.

The relationship between area covered with faeces and soil inorganic N was explored by running Spearman's correlations ($n=48$) between N concentrations ($\text{NH}_4^+\text{-N}$, $\text{NH}_3^-\text{-N}$, and N-inorg) at soil sampling dates 2 and 3, and LS means of area covered with faeces. This was done for the area covered with faeces at the end of sow batches 2 and 3, which occurred 21 and 34 days before soil sampling, and for sow batches

1 and 2, which occurred 113/117 and 154 days before soil sampling.

Additionally, the Spearman's correlation with soil inorganic N was tested for herbaceous cover model estimated as LS means for different days before the soil sampling date (0, 1, 2, 3, 4, 5, 6, 7, 14, 28, 42, 56 days) for the grass zones A and C.

Results

Site conditions

In 2023, the mean daily temperature approximated the 10-years average of 9.1 °C (9.2 °C in 2023 and 9.6 °C in 2024). Similarly, the annual precipitation of 1,250 and 1,218 mm in 2023 and 2024, respectively, was close to the 10-years average of 1,302 mm. The highest daily precipitation occurred in September 2023 (Fig. 5).

The topsoil (0–20 cm) texture was classified before pig introduction as coarse sand, following Danish soil classification (JB1; Breuning-Madsen et al. 2001) with 4.7% clay on average ($\text{SD}=0.3\%$). Samples contained 2.0% total C ($\text{SD}=0.3\%$) and 0.14% total N ($\text{SD}<0.1\%$) on average, which were higher in the tree zone compared to the grass zones by factor 1.3. The C/N ratio averaged 13.7 ($\text{SD}=0.3$).

Surface nitrogen balance

The total feed input was 490 kg N ha^{-1} for sows and 219 kg N ha^{-1} for piglets. Weaned, dead and otherwise lost piglets contributed 343, 9 and 12 kg N ha^{-1} to piglet weight gain, respectively. Including a sow weight loss of 28 kg N ha^{-1} , 373 kg N ha^{-1} was excreted. Including the further inputs of straw bedding, atmospheric deposition and biological fixation of 29, 11 and 22 kg N ha^{-1} , respectively, results in a surface N balance of 435 kg N ha^{-1} (Appendix 3, Table 4).

Sow elimination behaviour

The count of total eliminations ($n=141$) observed at three sow batches, in all sixteen paddocks and in three vegetation zones was influenced by the vegetation zone only ($\chi^2=12.6$, $\text{df}=2$, $p=0.002$) and not by tree management ($\chi^2=0.1$, $\text{df}=1$, $p=0.715$) nor resource position ($\chi^2<0.1$, $\text{df}=1$, $p=0.963$) or any

Fig. 5 Daily precipitation, evaporation and air temperature from 1 January 2022 to 31 December 2024 retained from national grid database of the Danish Meteorological Institute (DMI) for the experimental site

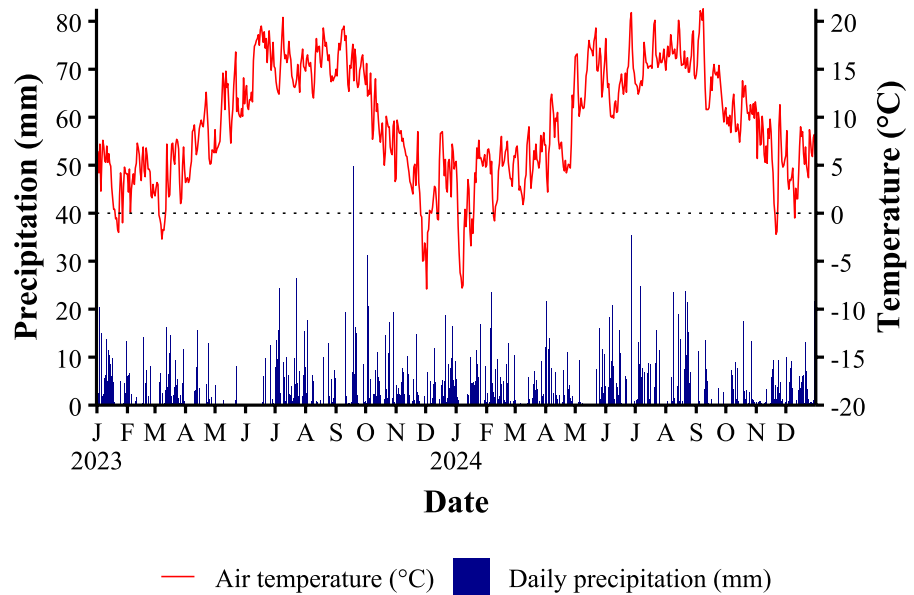


Table 1 Ratio of count of total eliminations comparing contrasted vegetation zones (grass zone A, poplar tree zone B and grass zone C) estimated as least-square means with indication of standard error (SE), z-ratio and adjusted p-values

Contrasted zones	Ratio	SE	z-ratio	p-value
A / B	0.252	0.115	−3.031	0.007
A / C	0.707	0.368	−0.667	0.783
B / C	2.805	1.121	2.580	0.027

of the interactions between them ($p > 0.2703$). Eliminations were observed more often in the tree zone B (31 times) than in the grass zones A (8 times) and C (11 times), respectively (Table 1). More than half of the observed urination events occurred in the tree zone (12 out of 22). Of the 28 defaecations, 19 were observed in the tree zone. During one observation day an average of 3.7 urinations ($SD = 2.5$) and 4.7 defaecations ($SD = 1.9$) were counted.

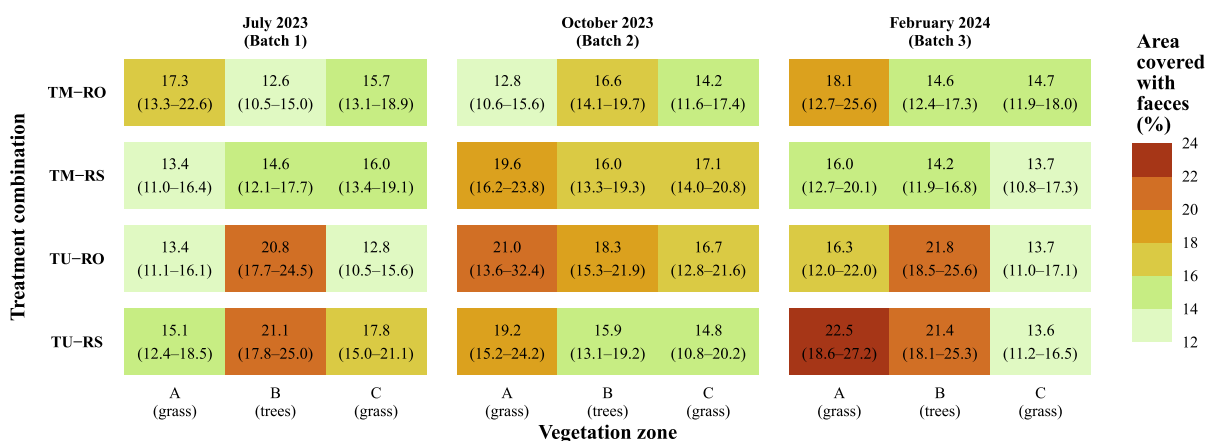


Fig. 6 The area covered with faeces (%) in paddock 5–12 estimated as least-square means and confidence interval (confidence level=0.95) for sampling occasion (date/batch), vegetation zone and treatment combination referring to the interac-

tion between tree management (managed (TM) or unmanaged (TU)) and resource position (on opposite sides (O) or on the same side of trees (S))

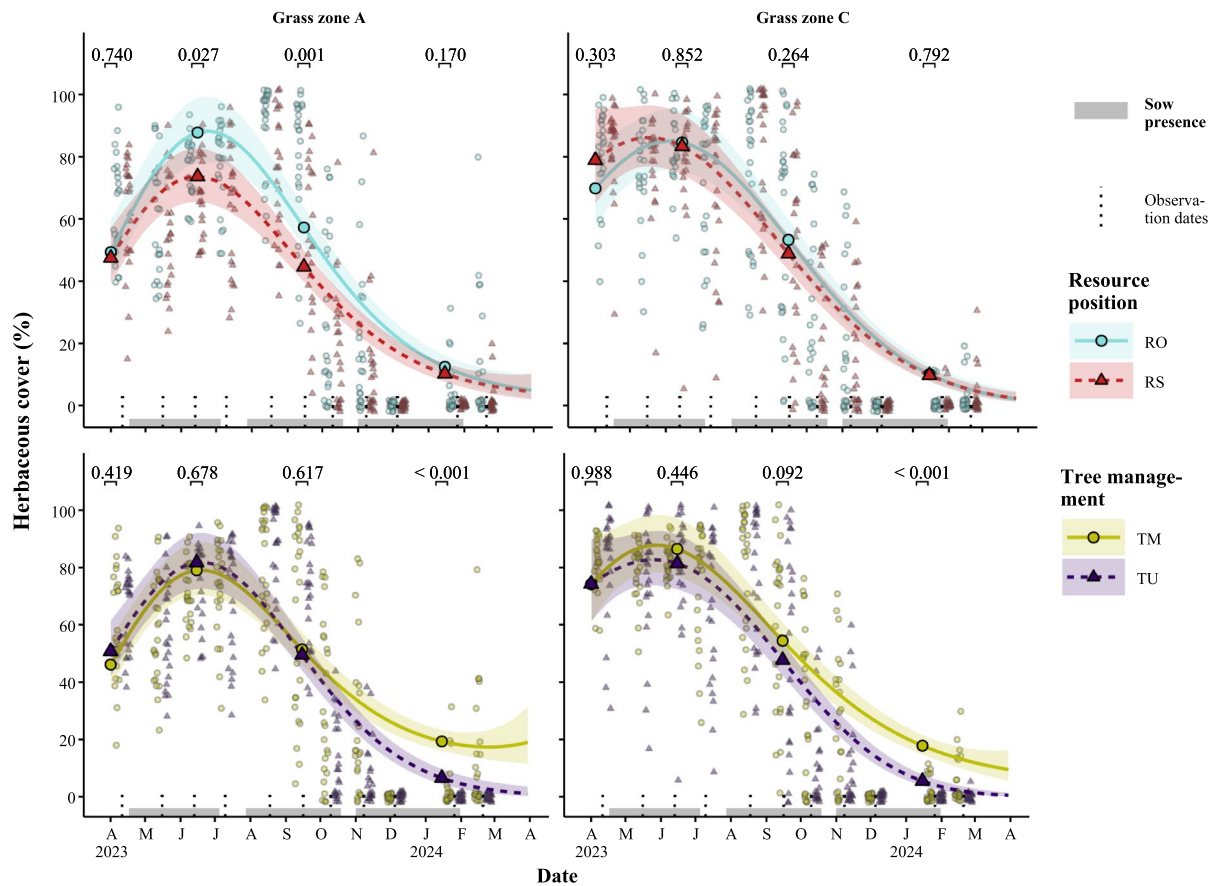


Fig. 7 Herbaceous cover referring to the fraction of paddock area covered by herbaceous vegetation in grass zones A and C as observed (small points) and as modelled by the generalized linear mixed model with date as a polynomial component (lines) and as least-square means (lines and large points) with confidence interval (shaded area; confidence level=0.95)

Area covered with faeces

A different pattern of the distribution of area covered with faeces across zones in paddocks 5 to 12 was observed in batch 1 compared to batches 2 and 3. However, of the only 0.9% of raw values ranging at or above 50% cover, most observations (190 out of 212 observations of 1 × 1 m) are found in the tree zone B. The LS means for each batch and treatment combination ranged from 12 to 22% (Fig. 6). The zone was the only main effect influencing the variability of the area covered with faeces ($\chi^2=7.13$, $df=2$, $p=0.028$), whereas tree management, resource position, zone and batch interacted with each other up to the fourth order (Appendix 3, Table 5), indicating

for each treatment combination of managed (TM) or unmanaged poplar trees (TU) and resources located on the same side (RS) or on opposite sides (RO) of the trees with indication of adjusted p-values (at the top) of post-hoc comparisons between treatment levels at chosen dates (large points)

that the area covered with faeces was non-consistent in its variability across the different factors.

When looking at post-hoc comparisons of zones, no consistent pattern emerged for paddocks with tree management (TM). If resources were placed on opposite sides of the trees (TM-RO), tree zone B had a lower cover compared to the grass zones A and C after batch 1 (ratio A/B=1.38, SE=0.18, $p=0.037$; ratio B/C=0.80, SE=0.07, $p=0.027$). After batch 2, on the contrary, the cover was higher in zone B compared to grass zone A (TM-RO: ratio A/B=0.77, SE=0.07, $p=0.007$). In paddocks with unmanaged trees (TU), the pattern was somewhat more consistent with relatively high cover in tree zone B, especially after batch 1 and 3. Regardless of resource position, zone B was

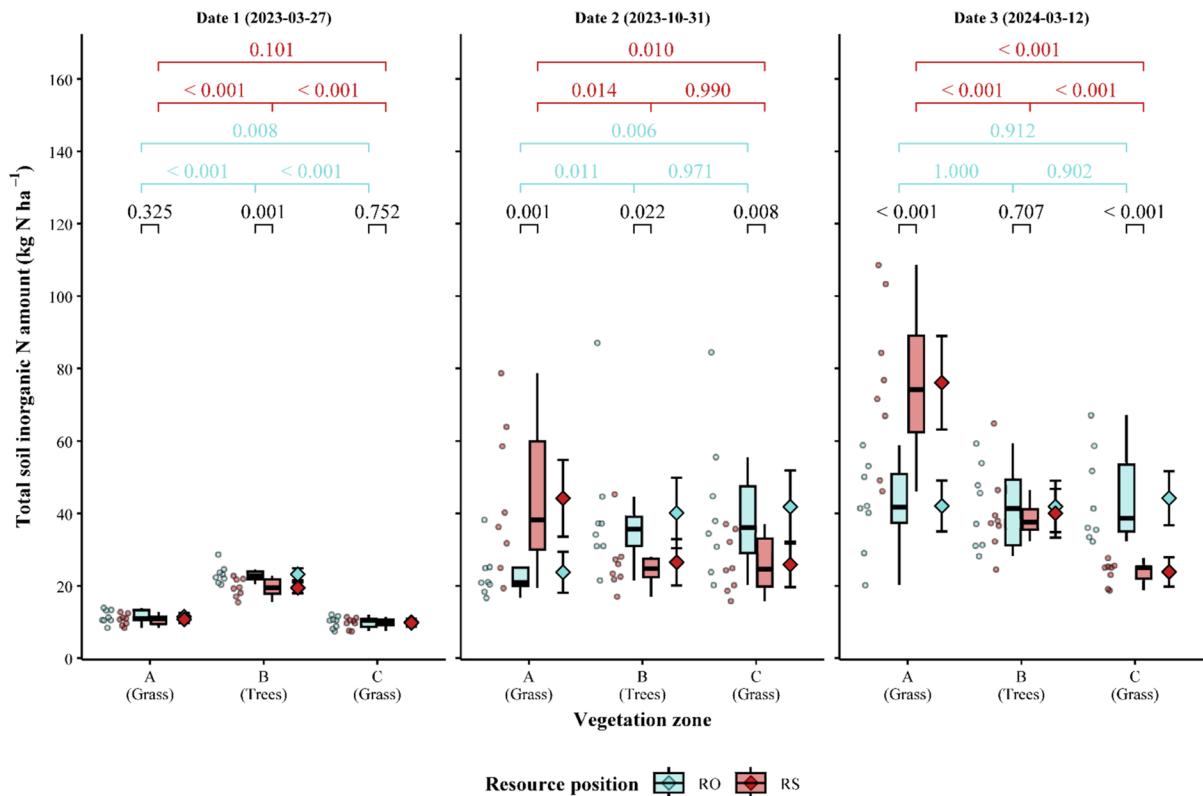


Fig. 8 Total soil inorganic-N amount ($\text{NO}_3^- \text{-N} + \text{NH}_4^+ \text{-N}$) as measured in the soil at the depth 0–20 cm (small points and boxplots) and as least-square means (large points) with confidence interval (error bars; confidence level=0.95) for resource position on opposite sides (RO; light blue) or on the same side (RS; red) of trees at each date (before sow introduction in March 2023 (1), after two batches of sows in October 2023 (2),

and after three batches of sows in March 2024 (3)) and vegetation zone (grass zone A, poplar tree zone B, and grass zone C) with indication of adjusted p-values for contrasted resource positions in each combination of zone and date (black) and for contrasted zones in each combination of resource position (indicated by colour) and date

covered with faeces to a greater extent than grass zone A after batch 1 (TU-RO: ratio A/B=0.64, SE=0.05, $p \leq 0.001$; TU-RS: ratio A/B=0.72, SE=0.7, $p \leq 0.001$) and grass zone C after batch 3 (TU-RO: ratio B/C=1.59, SE=0.16, $p \leq 0.001$; TU-RS: ratio B/C=1.58, SE=0.14, $p \leq 0.001$). However, zone A had a higher cover than zone C after batch 3, but only in paddocks where the resources were on the same side (TU-RS: ratio A/C=1.65, SE=0.16, $p \leq 0.001$).

Significant treatment differences emerged mainly in the tree zone B at batch 1 and 3 (Appendix 3, Table 6). Here, tree management reduced mean area covered with faeces from 21–22% (TU) to 13–15% (TM) of soil area covered. Differences between the resource positions within the same tree management were rarely significant.

Herbaceous cover

After an initial increase, the herbaceous cover showed a strong decline during the experimental period (Fig. 7). However, this strong influence of date as a third-degree polynomial ($\chi^2=578.68$, df=3, $p < 0.001$) also interacted with zone and tree management. Zone as the other significant main effect ($\chi^2=4.49$, df=1, $p=0.034$) interacted with resource position (Appendix 3, Table 7).

The LS mean estimates in spring (April 2023) were 46–51 and 70–79% cover in zones A and C, respectively. While this difference between zones was consistent in post-hoc comparison across treatments in April 2023 ($p \leq 0.001$ for all treatments), this was not the case during the rest of the year (Fig. 7). Only

with resources on the same side of the trees (RS), zone A continued to have a lower cover than zone C until summer (June 2023: ratio A/C=0.88, SE=0.06, $p=0.048$, Appendix 3, Table 7).

Placing resources on opposite sides of the trees (RO), increased herbaceous cover in zone A by 12 and 22 percentage points in June and September 2023, respectively, compared to resource position RS (June 2023: ratio RO/RS=1.19, SE=0.09, $p=0.027$; September 2023: ratio RO/RS=1.28, SE=0.10, $p=0.001$). No significant differences were observed in zone C (Appendix 3, Table 7).

Tree management (TM) allowed a higher herbaceous cover by 12 percentage points in both zone A and C in January 2024 (Zone A: ratio TM/TU=2.95, SE=0.06, $p\leq 0.001$; Zone B: ratio TM/TU=3.23, SE=0.05, $p\leq 0.001$). At this time, the cover was below 20% across zones and treatments. During the rest of the observation period, there were no considerable differences (Fig. 7; Appendix 3, Table 7).

Soil inorganic nitrogen

The concentrations of NO_3^- -N measured in the top-soil (0–20 cm) of each combination of zones, paddocks and dates ranged from 0.5 to 6.6 mg kg DM^{-1} , which is equivalent to a mass amount of 1.4 to 18.5 kg ha^{-1} compared to NH_4^+ -N ranging at 1.8–36.2 mg kg DM^{-1} corresponding to 5.0–101.4 kg ha^{-1} . The main effects on the parameters NO_3^- -N, NH_4^+ -N and N-inorg, were only significant for date and zone. However, tree management and resource position interacted significantly with zone and date, but not with each other (Appendix 3, Table 5).

During the time of pig presence, the amount of N-inorg increased with LS means of 14.1 (SE=0.3), 33.7 (SE=1.8) and 44.7 (SE=1.7) kg ha^{-1} initially in March 2023 (date 1), after 2 batches of sows in October 2023 (date 2) and after three batches of sows in March 2024 (date 3), respectively.

When comparing zones before pigs were introduced (date 1), soil N amount (NO_3^- , NH_4^+ and N-inorg) was significantly higher in the tree zone (B) than in grass zones (A and C) in all treatment combinations. In zone A, B and C, respectively, N-inorg was 11.1 (SE=0.4), 21.3 (SE=0.6) and 9.8 (SE=0.4) kg ha^{-1} averaged across treatments.

After pig presence, the differences between zones depended on the resource position, as the significant influence of the three-way interaction indicated (Appendix 3, Table 5). After two batches of sows (date 2) with resources located on the same side (RS), the LS mean of N-inorg was 44.2 kg ha^{-1} in zone A, where both hut and feeding trough were placed (Fig. 8; Appendix 3, Table 8). This represents increases of 67% and 71% compared to 26.5 and 25.9 kg ha^{-1} in zones B and C, respectively. Resources on opposite sides of the trees (RO) lead to the inverse pattern, i.e. with high amounts of inorganic N of 40.0 and 41.8 kg ha^{-1} in zone B and C, respectively, corresponding to 69 and 76% higher N amount than in zone A. In March 2024, after three batches of sows (date 3), a distinct pattern had developed. In zone A, the difference between resource positions had increased, with treatment RS having an inorganic N amount of 76.1 kg ha^{-1} compared to 42.1 kg ha^{-1} in RO. In zone B, there were no significant differences between resource positions, whereas zone C showed a significantly higher N-inorg amount of 44.2 kg ha^{-1} in treatment RO compared to 23.8 kg ha^{-1} in RS (Fig. 8; Appendix 3, Table 8, Table 9).

The tree management in interaction with the zone explained some of the variance of N-inorg (Appendix 3, Table 5). However, the direct comparison of treatments was not significant, but tree management showed the tendency to increase N-inorg amount across zones after batch 3 (estimate TM–TU=2.35, SE=1.20, $p=0.050$). Exploring the tendency of tree management to influence NO_3^- , an effect of tree management lowering the amount in grass zone A across dates was found (ratio TM/TU=0.75, SE=0.07, $p=0.004$).

Relationship between area covered with faeces, herbaceous cover and soil inorganic nitrogen

There was a moderate positive, monotonic relation between NO_3^- and area covered with faeces in the different batches (Spearman's correlation, $n=48$). This relation was similar for the area covered with faeces at the end of batch 2 and 3 ($r_s=0.27$, $p=0.07$), which was observed approximately 3 and 5 weeks before soil sampling date 2 and 3, respectively, and the area covered with faeces at the previous batch 1 and 2 observed approximately 16 and 22 weeks before soil sampling date 2 and 3 ($r_s=0.28$, $p=0.05$). No correlations were found between NH_4^+ and N-inorg, and

area covered with faeces ($|r_s| < 0.10$ and $p > 0.50$ for all correlations).

For herbaceous cover and soil inorganic N, the Spearman's correlation ($n=96$) was the highest for the herbaceous cover model estimated for each paddock for the exact day of soil sampling (-0.67 , -0.70 , and -0.72 for NO_3^- , NH_4^+ , and N-inorg, respectively. The larger the time difference between the cover estimate and the soil sampling, the lower the correlation index (Appendix 3, Table 10).

Discussion

Nitrogen deposition by free-range pigs

The average 2.3 urinations and 3.0 defaecations per sow per observation day found in this study were slightly lower than those reported by Jakobsen et al. (2018), who recorded an average of 2.8 urinations and 3.5 defaecations per lactating sow per day in comparable paddocks with two rows of poplar trees at one end and with comparable total daily observation time. This lower frequency persists despite starting observations earlier in the morning (7 am in batch 1 and 2, and 7.40 am in batch 3) to capture expected early morning eliminations close to the hut, as suggested by Jakobsen et al. (2018). When comparing to 24-h recordings of indoor lactating sows, which showed 3.1–3.4 urinations and 1.5–4.5 defaecations per sow per day depending on the stage of lactation and feeding strategy (Andersen and Pedersen 2011; Tabeling et al. 2003), the frequencies observed in this study suggest that some eliminations, possibly those occurring in the very early morning and during the night, were missed.

Total sow elimination counts were higher in the tree zone (approx. one third of the paddock area) than in the grass zones. Urination and defecation followed a similar pattern. This indicates that sows used the trees differently than grass, either for activity in general or eliminations specifically. Compared to Jakobsen et al. (2018), which observed 7% of urinations and 19% of defaecations in the tree zone covering 18% of the paddock area, locating the trees in the centre of the paddock seems to have increased the proportion of urine and faeces deposited in the tree zone. However, piglet behaviour was not observed, although their eliminations might also influence the soil N hotspots.

The total counts of observed sow elimination behaviour were relatively low, which limited the ability to detect treatment effects. Not only longer observation time windows but also more observation days would enhance the robustness of the data. For future studies, it is important to consider the development of alternative indicators of sow elimination behaviour, as direct observations are time-consuming, and video observations are challenged by the presence of trees. Higher count observations would also allow the differentiation of urinations and defaecations, which is highly relevant to short-term N availability, since about 70% of N excreted by lactating sows is contained in the urine (Dourmad et al. 1999).

The visual scoring of faeces covered area only partially supports the findings from the sow behavioural observations. This may be partly explained by inclusion of piglet faeces in the visual scoring although piglet elimination site preferences are in general expected to reflect that of sows, except during the first few days in life (Andersen et al., 2020). Also, when considering faeces hotspots, defined as more than 50% cover within the observed 1×1 m, these were most often (90% of hotspots) located within the tree zone. Additionally, LS means of all area covered with faeces recordings show that the covered area was larger in the tree zone compared to grass zones, especially after batches 1 and 3, but only in unmanaged trees (TU; Fig. 6). This difference between tree zones with and without management might be a result of high precipitation a few days before these two sampling occasions. Field reports from the data collector suggested that rainfall washed faeces between WC. Therefore, area covered with faeces under managed trees (TM) might have been underestimated compared to more accurate estimates for the soil surface under unmanaged trees (TU) and in grass zones. Additionally, in October 2023, after batch 2, leaf litter covered the soil surface in tree zones, which hampered the estimation. Hence, it can be assumed that the paddocks with unmanaged trees (TU) without autumn leaf litter (after batch 1 and 3) represent the most robust observations. All in all, these factors impeding observations explain the low correlation with soil inorganic N. Additionally, the organic faeces N must be mineralized before it can influence soil inorganic N.

Nitrogen uptake by herbaceous vegetation

The strongly decreasing herbaceous cover in autumn with sows present (Fig. 7) suggests that equipping the sows with nose-rings does not suffice to inhibit vegetation destruction, as reported before (Eriksen et al. 2006). Soil inorganic N and herbaceous cover were strongly correlated, with the strongest association observed when herbaceous cover was estimated on the same day as soil sampling, rather than from 3 days to 6 weeks prior. This likely reflects the absorption of inorganic N by the current vegetation, with potential nitrogen losses through leaching increasing as the time interval between vegetation estimation and soil sampling widens, especially under conditions of sufficient precipitation.

Soil inorganic nitrogen in farrowing paddocks

The NO_3^- amount of $14.1 \text{ kg N ha}^{-1}$ on average before pig introduction was comparable to $14\text{--}16 \text{ kg N ha}^{-1}$ found for comparable paddocks (Eriksen et al. 2002; Kongsted et al. 2020). After two and three batches of lactating sows, respectively, inorganic N amounts were 33.7 and $44.7 \text{ kg N ha}^{-1}$ across treatments and zones, with peaks of 87.1 and $108.6 \text{ kg N ha}^{-1}$. These N levels are much lower than amounts reported from other pig systems without (Eriksen et al. 2002) or with trees integrated and with corresponding surface N balances (Jørgensen et al. 2018), where hotspots reached up to 500 kg N ha^{-1} (Jakobsen et al. 2018). This could be due to deviations from the current experiment setup such as larger sampling depth ($0\text{--}40 \text{ cm}$ in Eriksen et al. (2002); $0\text{--}50 \text{ cm}$ in Jakobsen et al. (2018)), smaller sampling distance without pooling which increases variation ($1\text{--}4 \text{ m}$ in Jakobsen et al. (2018)), examination of growing pigs without nose-rings instead of lactating sows with nose-rings (Jørgensen et al. 2018). Another factor decreasing inorganic N in the topsoil is the high rainfall in August and late September 2023, causing high percolation and thereby downward transport of NO_3^- . Finally, as the piglets were able to roam freely, similar to the conditions described by Jakobsen et al. (2018), a proportion of their excretions may have been deposited outside the experimental area and thereby not reflected in the soil N levels within the experimental paddocks.

The findings of initially higher NO_3^- -N and NH_4^+ concentrations under trees compared to grass-clover ley (Fig. 8) were likely connected to leaf litter. Thevathasan and Gordon (1997) found that 50 kg N ha^{-1}

year^{-1} were contributed by leaf litter collected at $0\text{--}2.5 \text{ m}$ distance to rows of 7 m high poplar trees in an alley-cropping system, which supposedly increased nitrification rates. This contribution fits the current finding that total C and N concentration in the tree zone were higher by factor 1.3 compared to the grass zones. Additionally, an effect of former pig presence is possible, since the average of $21.3 \text{ kg N ha}^{-1}$ (inorganic N) is higher than $17.0 \text{ kg N ha}^{-1}$ observed in the depth $0\text{--}50 \text{ cm}$ under trees without pigs in March in Jakobsen et al. (2018). Another influence is the harvest of spring barley every two years in the grass zones, which is estimated to remove 65 kg N ha^{-1} , including straw, each time (Kongsted et al. 2020), whereas no biomass was removed from the tree area.

Differences in soil inorganic N between vegetation zones after the presence of sows interact with treatment effects. In addition to influences of the vegetation on animal behaviour (compare Sect. "Nitrogen deposition by free-range pigs"), poplars affect soil N differently than pasture plants. Poplar trees have higher fine root biomass at the depth $40\text{--}60 \text{ cm}$ (Fortier et al. 2013), can accumulate more N (Fortier et al. 2015, 2010), and their N uptake remains higher during the non-growing season compared to herbaceous vegetation (Ma et al. 2021). However, near the soil surface ($0\text{--}20 \text{ cm}$), the fine root biomass is higher in herbaceous vegetation (Fortier et al. 2013).

Effect of resource position on nitrogen cycling

A higher soil inorganic N amount in the tree zone of RO was anticipated when the hut was located on its opposite side (RO) compared to on the same side (RS). Indeed, locating the hut on the opposite side of the trees, in zone C, at first moved the zones of high soil inorganic N from zone A with the feeding station to zone B (trees) and C (grass east). Later, it equalled out the zone differences with soil inorganic N levels in zone A with the feeding station remaining high. This aligns with earlier findings of high inorganic N found close to feeding areas (Jakobsen et al. 2018) which can be caused by feed waste. However, feed waste is expected to be low due to the farmer's effort to reduce it, such as a lid on the feeding trough. The two resources being placed on the same side clearly increased N accumulation on this side, zone A, so that N availability will exceed plant demand and is likely to be leached. Overall, the results indicate that

separating the resources hut and feeding trough across the tree zone may contribute to a more even distribution of inorganic N within the paddock.

This effect of resource position was not evident in elimination behaviour or in the area covered with faeces (Sect. "Nitrogen deposition by free-range pigs"). Most eliminations and faecal hotspots were observed in the tree zone, but this did not elevate soil inorganic N above the levels found in the grass zones. This may partly be explained by N exceeding the immobilisation capacity and being lost from the topsoil during winter. At moderate soil temperatures, poplar trees with most of their fine roots in the depth 0–20 cm (Fortier et al. 2013) could have absorbed some inorganic N even during the non-growing season (Ma et al. 2021). However, in coarse sandy soil, any remaining soluble N in the form of NO_3^- is likely to be leached from the topsoil due to the high rainfall typical of Danish winters (Eriksen 2001; Eriksen et al. 2002).

Soil organic N amount in the grass zones was influenced by the resource position and strongly negatively correlated to herbaceous cover, which showed the opposite reaction to the resource position. In the grass zone with the feed and hut (zone A), herbaceous cover in June and September in 2023 was increased by 12–22 percentage points by moving the hut to the other side of the trees (zone C), where no difference was observed. The preserved vegetation in zone A thus contributed to reducing available N amount in the soil, as it was observed earlier by Eriksen et al. (2006), supposedly by N uptake and restrained N mineralisation due to evaporation lowering soil moisture.

Overall, the importance of resource position aligns with earlier findings that regular moving of resources supports a more even distribution of nutrients (Eriksen and Kristensen 2001; Jakobsen et al. 2018). Apparently, the location of the hut determined the location of the highest soil inorganic N amount, which was partly mediated by the herbaceous cover.

Effect of tree management on nitrogen cycling and herbaceous cover

By combining tree cutting and WC surface mulching, the tree management could influence the N dynamics in several ways. Tree cutting might change the micro-environment regarding air temperature, humidity, and wind. Regrowth of shoots might amplify N demand, although reduced evaporation might limit the uptake.

Applying WC as surface mulch adds organic C, might reduce and balance the topsoil temperature, and helps to retain soil moisture, which are factors influencing N net mineralisation (Yansheng et al. 2020).

However, in this sampling of 16 paddocks with non-randomized location of tree management treatments, no significant influence on soil inorganic N was observed. Tendencies showed that managing the trees might increase N-inorg amount across zones with no indication for increased N demand of managed trees. This could be connected to rust (*Melampsora* sp.) having infected 90% of leaves of the managed trees at the end of the growing season in 2023.

Furthermore, the expected effect of WC in reducing soil inorganic N through microbial immobilisation (Yansheng et al. 2020) was not evident at the scale of this study. In contrast, Sønsteby et al. (2004) observed up to 25% reduction in soil NO_3^- -N using spruce bark mulch, at high N fertilisation level (120 kg N ha^{-1}), likely due to its higher C/N ratio of 135. In comparison, WC (pecan) with a C/N ratio of 30, similar to the current study's ratio of 33, did not immobilise N even when incorporated (Tahboub et al. 2007), suggesting this ratio is insufficient to trigger additional N for decomposition. This and the small particle size explain why applying hardwood sawdust with a C/N ratio of about 500 reduced net mineralisation in Averett et al. (2004). Regarding the size of WC, the results from a pot experiment suggested that WC smaller than 1.6 mm promoted stronger N immobilisation and soil C stabilisation within 8 months, compared to WC up to 12.7 cm (Gao et al. 2024). Hence, it is likely that the decomposition of the relatively large WC used in this experiment (1–5 cm) was too slow to significantly reduce the high soil N levels. This is supported by the low topsoil C/N ratio of 14 measured in March 2023 one year after application of $20 \text{ t organic C ha}^{-1}$ from WC with an estimated C/N ratio of 33. Overall, the lack of incorporation, the relatively large size, and the low C/N ratio of the WC limited the C decomposition and thus N immobilisation.

Tree management increased herbaceous cover in both grass zones by 12 percentage points in January 2024. The higher herbaceous cover under tree management may be linked to shading effects, though this influence should be minimal in grass zone A due to its position in the south-east of the trees. Interestingly, the variation in herbaceous cover during winter when the tree canopies are leafless suggests other factors may also be at play. The higher herbaceous

cover under tree management is more likely a result of changes in foraging behaviour, since pigs are known to change their diet seasonally. Rooting activity peaks in autumn and winter (Calosi et al. 2024), and WC being an attractive rooting material (Jensen and Pedersen 2007) may help reduce vegetation destruction. Bornett et al. (2003) demonstrated that vegetation can be protected by providing a designated rooting area with mushroom compost. In the present study, piglets, unlike sows, were not nose-ringed and remained in the paddock until reaching approximately 20 kg, suggesting they contributed substantially to rooting activity.

The tree management treatment represented a management strategy feasible for farmers. The selling of WC for energy purposes is missed out, but the relatively small-sized paddock areas are not always profitable to be harvested for selling anyway. However, the resulting confoundment of tree cutting and WC mulching did not allow their separate evaluation, which is a suggestion for future research. In addition, the effect of smaller-sized WC incorporated into the soil instead of surface mulching should be investigated.

Conclusion

This on-farm experiment evaluated a novel sow paddock design incorporating centrally located poplar trees in a pasture-based system. As hypothesized, separating key resources by placing the hut and feeding trough on opposite sides of the trees altered the spatial distribution of soil inorganic N. This arrangement resulted in a more even distribution of topsoil inorganic N across vegetation zones compared to placing resources on the same side of the trees. After three sow batches, inorganic N amounts were similar among zones in the separated-resource design, whereas resource location on the same side produced pronounced accumulation in the grass zone containing both feed and hut. During summer and autumn, separating resources helped to maintain herbaceous cover in the zone with the feeding trough. However, differences in soil inorganic N between resource positions could not be fully explained by herbaceous cover, and elimination behaviour observations were probably too few to show treatment differences. Tree management in terms of cutting trees at 2 m height with surface application of coarse wood chips did not influence soil inorganic N despite expectations of reduced N availability through

carbon inputs and tree regrowth. Limited soil incorporation of woodchips, their size, and C/N ratio likely constrained these effects. Nevertheless, increased herbaceous cover following winter suggests that wood chips redirected pig rooting behaviour away from the grass zones. Overall, the results demonstrate that a paddock system with centrally located poplar trees combined with strategic placement of the hut and feeding can improve the spatial distribution of soil inorganic N through pig rooting and elimination behaviour, thereby potentially reducing the risk of nitrate leaching in pasture-based systems.

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Author contributions Magdalena Ullfors: Validation, Methodology, Formal analysis, Data curation, Writing—Original Draft, Writing—Review and Editing, Visualization Kiril Manevski: Validation, Investigation, Writing—Review and Editing Uffe Jørgensen: Conceptualization, Validation, Methodology, Writing—Review & Editing, Funding acquisition Maarit Mäenpää: Validation, Methodology, Formal analysis, Writing—Review and Editing Søren Ugilt Larsen: Validation, Writing—Review & Editing, Funding acquisition Martin Jensen: Validation, Methodology, Writing—Review and Editing, Funding acquisition Anne Grete Kongsted: Conceptualization, Validation, Methodology, Writing—Original Draft, Writing—Review and Editing, Project administration, Funding acquisition.

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Data availability Data will be available on dryad.org.

Declarations

Conflicts of interest The authors declare no conflict of interest.

Ethical approval All animals included in the study were treated according to the regulations of organic agriculture in Denmark (Landbrugsstyrelsen 2025).

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Appendix 1

See Table 2

Table 2 Details on the generalized linear mixed models used to explore effects of vegetation zone ("zone", grass zones A and C, and poplar tree zone B), sampling occasion (sow batch or date), and treatment factors (two tree management treatments and two resource position treatments) on elimination behaviour, area covered with faeces, herbaceous

cover and three forms of soil inorganic nitrogen (N), which are nitrate (NO_3^-), ammonium (NH_4^+), and total inorganic N ($\text{NO}_3^- + \text{NH}_4^+$; N-inorg), fitted using the glmmTMB R package (v1.1.10, Brooks et al. 2017). The interactions refer to all possible combinations of main effects with indication of their number

Dependent variable	Elimination behaviour	Area covered with faeces	Herbaceous cover	Soil inorganic N		
	Count of total eliminations (n)	Fraction of soil covered with faeces (m^2 covered m^{-2})	Fraction of soil covered by vegetation (m^2 covered m^{-2})	NO_3^- -N concentration (mg N kg DM^{-1})	NH_4^+ -N concentration (mg N kg DM^{-1})	N-inorg concentration (mg N kg DM^{-1})
No. of observations (n)	141	22 680	1 056	96	96	96
Fixed effects:	Zone (A, B, C),	Sow batch (1–3),	Date (as a third-	Date (1–3),	Date (1–3),	Date (1–3),
Main effects	Tree management,	Zone (A, B, C),	degree polynomial term on 11	Zone (A, B, C),	Zone (A, B, C),	Zone (A, B, C),
	Resource position	Tree management,	dates),	Tree management,	Tree management,	Tree management,
		Resource position	Vegetation zone (A, C),	Resource position	Resource position	Resource position
			Tree management,			
			Resource position			
2-way interactions	Included (4)	Included (6)	Included (6)	Included (6)	Included (6)	Included (6)
3-way interactions	Included (1)	Included (4)	Not included	Included (4)	Included (4)	Included (4)
4-way interactions	-	Included (1)	Not included	Included (1)	Included (1)	Included (1)
Random effects	Paddock (1–16) nested in:	Paddock (5–12),	Interaction of	Paddock (1–16),	Paddock (1–16),	Paddock (1–16),
	o Sow batch (1–3)	Row (1–63),	paddock (1–16)	Paddock nested in:	Paddock nested in:	Paddock nested in:
	o Tree management	Interaction of	and subzone (1–3)	o Tree management	o Tree management	o Tree management
	o Resource position	paddock and column (1–15)		o Resource position	o Resource position	o Resource position
Probability distribution	Negative binomial	Zero-inflated gamma	Zero-inflated gamma	Gamma	Gamma	Gamma
Link function	Log	Log	Log	Log	Identity	Identity
Additions	Optimizer function "BFGS"	Zero-inflation function with zone as grouping variable	Zero-inflation function with zone as grouping variable	Dispersal function with date as grouping variable	Dispersal function with date as grouping variable	Dispersal function with date as grouping variable

Appendix 2

See Table 3 and Fig. 9

Table 3 The size class distribution and average weight of poplar chip particles derived from sorting and counting the particles of 85 g of bark and 818 g of wood before weighing the

total weight of each size class. The sorting size class refers to the shortest side of particles when lying on the largest side, which corresponds to the sieve opening size to pass through

Size class (cm)	Distribution of particles (weight % of total)		Average weight of particles (g)	
	Bark	Wood	Bark	Wood
0–0.5	0.6	0.6	0.1	0.0
0.5–1	3.1	5.5	0.4	0.3
1–2	7.1	19.2	1.0	1.3
2–3	4.7	25.1	2.5	3.5
3–5	0.8	27.2	4.1	9.1
> 5	0.0	6.2	–	30.1



Fig. 9 Placing of wood chips under poplar trees cut down to 2 m height. Photo: Kristine Vigh Riis

Appendix 3

See Table 4, 5, 6, 7, 8, 9 and 10

Table 4 Nitrogen surface balance for the 16 farrowing paddocks with poplar trees in the centre

Component	kg ha ⁻¹
Inputs	799.2
- Feed, sow	489.8
- Feed, piglets	219.4
- Sow weight loss	27.9
- Bedding	29.2
- Atm. deposition	11.0
- Fixation by grass-clover	21.9
Outputs	363.9
- Weaned piglets	343.2
- Dead piglets	9.0
- Disappeared piglets	11.7
Surface N balance	435.3

Table 5 Effects of main factors of date, zone, tree management, resource position, and their interactions on area covered with faeces (n = 22 680), herbaceous cover (n = 1 056), and concentrations of nitrate-N (NO_3^-), ammonium-N (NH_4^+) and total inorganic N ($\text{NO}_3^- + \text{NH}_4^+$; N-inorg) in topsoil (0–20 cm; n = 144) tested by a Wald Chi-Square test (type II). Some interactions were not included in the model of herbaceous cover

Variable	Area covered with faeces			Herbaceous cover			NO_3^- -N			NH_4^+ -N			N-inorg		
	χ^2	df	p	χ^2	df	p	χ^2	df	p	χ^2	df	p	χ^2	df	p
Date (D)	4.80	2	0.090	587.68	3	<0.001	415.38	2	<0.001	255.38	2	<0.001	381.18	2	<0.001
Zone (Z)	7.13	2	0.028	4.49	1	0.034	280.83	2	<0.001	414.94	2	<0.001	423.32	2	<0.001
Tree manage- ment (T)	2.63	1	0.105	1.32	1	0.250	3.09	1	0.079	0.00	1	0.988	0.54	1	0.462
Resource position (R)	0.25	1	0.620	3.30	1	0.069	1.71	1	0.191	1.31	1	0.252	1.95	1	0.163
Z×T	36.72	2	<0.001	1.93	1	0.165	4.80	2	0.091	12.54	2	0.002	6.72	2	0.035
Z×R	2.20	2	0.332	5.92	1	0.015	11.82	2	0.003	7.27	2	0.026	9.69	2	0.008
D×Z	18.59	4	<0.001	17.30	3	<0.001	10.68	4	0.030	40.00	4	<0.001	31.88	4	<0.001
D×T	1.11	2	0.574	33.20	3	<0.001	4.47	2	0.107	1.75	2	0.417	1.82	2	0.403
D×R	0.69	2	0.710	3.87	3	0.276	1.98	2	0.372	2.21	2	0.332	1.38	2	0.502
T×R	0.01	1	0.907	0.14	1	0.712	1.32	1	0.250	0.02	1	0.892	0.00	1	0.999
Z×T×R	2.11	2	0.348	-	-	-	0.69	2	0.710	0.99	2	0.611	0.42	2	0.810
D×Z×P	25.65	4	<0.001	-	-	-	8.45	4	0.076	0.59	4	0.964	0.84	4	0.932
D×Z×R	9.61	4	0.048	-	-	-	8.12	4	0.087	53.70	4	<0.001	58.92	4	<0.001
D×T×R	15.60	2	<0.001	-	-	-	4.59	2	0.101	1.05	2	0.590	0.30	2	0.862
D×Z×T×R	13.35	4	0.010	-	-	-	1.23	4	0.874	7.08	4	0.132	7.70	4	0.103

Table 6 Ratio between area covered with faeces estimated as least-square means for contrasted treatments (Treat. 1/treat. 2), which combine tree management (managed (TM) or unmanaged (TU) and resource position (on opposite sides (RO) or on the same side of trees (RS)) with indication of levels of batch and vegetation zone (zone) as well as standard error (SE), and p-value

Zone	Batch 1				Batch 2			Batch 3			
	Treat. 1	Treat. 2	Ratio		p-value	Ratio		p-value	Ratio		p-value
			Ratio	SE		Ratio	SE		Ratio	SE	
A	TM-RO	TM-RS	1.29	0.21	0.389	0.65	0.09	0.007	1.13	0.24	0.936
A	TM-RO	TU-RO	1.29	0.21	0.366	0.61	0.15	0.171	1.11	0.26	0.970
A	TM-RO	TU-RS	1.14	0.19	0.845	0.67	0.10	0.037	0.80	0.16	0.684
A	TM-RS	TU-RS	0.88	0.12	0.809	1.03	0.15	0.998	0.71	0.10	0.081
A	TU-RO	TM-RS	1.00	0.13	1.000	1.07	0.25	0.993	1.02	0.19	1.000
A	TU-RO	TU-RS	0.88	0.12	0.792	1.09	0.27	0.983	0.72	0.13	0.255
B	TM-RO	TM-RS	0.86	0.11	0.642	1.04	0.12	0.989	1.03	0.12	0.992
B	TM-RO	TU-RO	0.60	0.07	<0.001	0.91	0.11	0.851	0.67	0.07	0.002
B	TM-RO	TU-RS	0.60	0.07	<0.001	1.05	0.13	0.980	0.68	0.08	0.004
B	TM-RS	TU-RS	0.69	0.09	0.016	1.01	0.13	1.000	0.66	0.08	0.002
B	TU-RO	TM-RS	1.42	0.17	0.019	1.14	0.14	0.702	1.54	0.17	<0.001
B	TU-RO	TU-RS	0.99	0.11	0.999	1.15	0.14	0.671	1.02	0.11	0.999
C	TM-RO	TM-RS	0.98	0.12	0.999	0.83	0.11	0.546	1.07	0.17	0.971
C	TM-RO	TU-RO	1.23	0.16	0.384	0.85	0.14	0.762	1.07	0.16	0.907
C	TM-RO	TU-RS	0.88	0.11	0.744	0.96	0.18	0.997	1.08	0.15	0.951
C	TM-RS	TU-RS	0.90	0.11	0.801	1.15	0.21	0.864	1.01	0.15	1.000
C	TU-RO	TM-RS	0.80	0.10	0.301	0.98	0.16	0.999	1.00	0.16	1.000
C	TU-RO	TU-RS	0.72	0.09	0.044	1.13	0.23	0.937	1.01	0.14	1.000

Table 7 Marginal means of herbaceous cover (% surface area covered by herbaceous vegetation) estimated as least-square means (LS means, μ) with confidence intervals (CI) and their post-hoc comparisons presented as ratio with standard error (SE), and adjusted p-values (p) for different dates, zones (grass zones A and C) and treatment factors including managed (TM) or unmanaged poplar trees (TU) and resources located on the same (RS) or opposite sides of the trees (RO), as retained from the generalized linear mixed model

Date	Zone	LS means		Comparing treat				Comparing zones A/C			
		Treatment RO	Treatment RS	RO/RS		in RO		in RS		in TU	
		μ (CI)	μ (CI)	Ratio	SE	Ratio	SE	Ratio	SE	Ratio	SE
01 Apr. 2023	A	49.4 (40.9–59.7)	47.5 (39.4–57.3)	1.04	0.12	0.740	0.08	0.60	0.07	0.60	0.07
	C	69.8 (57.8–84.4)	78.9 (65.0–95.7)	0.89	0.10	0.303					
15 Jun. 2023	A	87.8 (78.0–98.8)	73.6 (65.4–82.8)	1.19	0.09	0.027	0.07	0.88	0.06	0.88	0.06
	C	84.6 (75.2–95.1)	83.4 (74.2–93.7)	1.01	0.08	0.852					
15 Sep. 2023	A	57.2 (51.3–63.8)	44.7 (39.9–50.0)	1.28	0.10	0.001	0.06	0.91	0.05	0.91	0.05
	C	53.3 (47.6–59.7)	48.9 (43.6–54.8)	1.09	0.08	0.264					
15 Jan. 2024	A	12.5 (9.7–16.2)	10.2 (7.7–13.6)	1.22	0.18	0.170	0.15	1.05	0.14	1.05	0.14
	C	10.1 (7.7–13.4)	9.8 (7.9–12.1)	1.04	0.15	0.792					
Date	Zone	Treatment TM		Treatment TU		TM/TU		in TM		in TU	
		μ (CI)	μ (CI)	Ratio	SE	p	SE	Ratio	SE	Ratio	SE
01 Apr. 2023	A	46.2 (38.3–55.7)	50.8 (42.0–61.5)	0.91	0.11	0.419	0.07	0.62	0.07	0.68	0.07
	C	74.1 (61.3–89.7)	74.3 (61.2–90.1)	1.00	0.12	0.988					
15 Jun. 2023	A	79.1 (70.3–88.9)	81.8 (72.5–92.2)	0.97	0.08	0.678	0.06	1.00	0.06	1.00	0.06
	C	86.6 (77.0–97.2)	81.5 (72.4–91.6)	1.06	0.08	0.446					

Table 7 (continued)

Date	Zone	LS means		Comparing treat			Comparing zones A/C					
		Treatment RO	Treatment RS	RO/RS		p	in RO			in RS		
		μ (CI)	μ (CI)	Ratio	SE		Ratio	SE	p	Ratio	SE	p
15 Sep. 2023	A	51.5 (46.3–57.4)	49.6 (44.3–55.6)	1.04	0.08	0.617	0.94	0.05	0.325	1.04	0.06	0.540
	C	54.5 (48.9–60.9)	47.8 (42.5–53.8)	1.14	0.09	0.092						
15 Jan. 2024	A	19.4 (16.1–23.4)	6.6 (4.5–9.7)	2.95	0.60	≤ 0.001	1.08	0.13	0.503	1.19	0.16	0.192
	C	17.9 (14.8–21.6)	5.5 (3.9–7.8)	3.23	0.63	≤ 0.001						

Table 8 Least-square means (LS means, μ) with confidence intervals (confidence level=0.95) for nitrate (NO_3^-), ammonium (NH_4^+) and total inorganic N ($\text{NO}_3^- + \text{NH}_4^+ + \text{N}$; N-inorg) in the soil at the depth 0–20 cm in the unit kg N ha^{-1} for treatment factors including managed (TM) or unmanaged poplar trees (TU) and resources located on the same (RS) or opposite sides of the trees (RO), at each date (before sowing introduction in March 2023 (1), after two batches of sows in October 2023 (2), and after three batches of sows in March 2024 (3)) and each vegetation zone (grass zone A, poplar tree zone B, and grass zone C) with indication of standard error (SE)

Date	Zone	$\text{NO}_3^- + \text{N}$		$\text{NH}_4^+ + \text{N}$		N-inorg	
		RS	RO	RS	RO	RS	RO
1	A	3.0 (2.6–3.4)	3.3 (2.8–3.8)	7.7 (7.0–8.4)	8.2 (7.5–8.9)	10.7 (9.7–11.7)	11.4 (10.4–12.5)
	B	5.0 (4.4–5.8)	6.5 (5.6–7.5)	14.3 (13.3–15.4)	16.7 (15.5–17.9)	19.4 (18.0–20.9)	23.2 (21.5–24.9)
	C	2.7 (2.3–3.1)	2.9 (2.5–3.4)	7.0 (6.4–7.7)	7.0 (6.4–7.6)	9.7 (8.7–10.7)	10.0 (9.0–10.9)
2	A	6.1 (4.9–7.7)	4.4 (3.5–5.6)	37.4 (27.3–47.6)	19.1 (13.9–24.2)	44.2 (33.6–54.8)	23.7 (18.1–29.4)
	B	7.1 (5.7–9.0)	11.3 (9.0–14.2)	19.0 (13.7–24.4)	28.8 (20.9–36.7)	26.5 (20.1–32.9)	40.1 (30.4–49.9)
	C	5.9 (4.7–7.4)	5.6 (4.5–7.1)	19.9 (14.5–25.3)	36.0 (26.2–45.9)	25.9 (19.6–32.1)	41.8 (31.7–51.9)
3	A	6.5 (5.4–7.8)	5.8 (4.8–7.0)	69.3 (55.0–83.6)	36.1 (28.8–43.4)	76.1 (63.2–89.0)	42.1 (35.0–49.1)
	B	10.8 (9.0–13.0)	12.6 (10.5–15.2)	29.2 (23.2–35.1)	29.1 (23.1–35.1)	40.1 (33.3–46.8)	41.9 (34.8–49.0)
	C	6.7 (5.5–8.1)	7.5 (6.2–9.0)	16.9 (13.5–20.4)	36.7 (29.2–44.2)	23.8 (19.8–27.9)	44.2 (36.7–51.7)
Date	Zone	TM	TU	TM	TU	TM	TU
1	A	2.9 (2.5–3.4)	3.3 (2.9–3.8)	7.4 (6.7–8.1)	8.5 (7.8–9.2)	10.4 (9.4–11.4)	11.8 (10.7–12.9)
	B	5.0 (4.4–5.8)	6.4 (5.6–7.5)	15.6 (14.5–16.8)	15.4 (14.3–16.5)	20.7 (19.2–22.3)	21.9 (20.2–23.5)
	C	2.6 (2.3–3.0)	3.0 (2.6–3.4)	7.4 (6.7–8.0)	6.7 (6.0–7.3)	10.0 (9.0–11.0)	9.7 (8.7–10.7)
2	A	4.1 (3.3–5.2)	6.6 (5.3–8.3)	28.0 (19.9–36.1)	28.5 (20.5–36.5)	32.2 (24.1–40.4)	35.7 (26.8–44.5)
	B	8.2 (6.6–10.4)	9.8 (7.8–12.3)	24.6 (17.9–31.2)	23.3 (16.5–30.1)	33.4 (25.4–41.5)	33.2 (24.8–41.6)
	C	5.7 (4.6–7.2)	5.8 (4.7–7.3)	30.9 (22.0–39.7)	25.1 (18.1–32.0)	36.5 (27.3–45.6)	31.2 (23.7–38.8)
3	A	5.4 (4.5–6.5)	7.0 (5.8–8.4)	59.3 (46.4–72.3)	46.1 (36.6–55.5)	64.8 (53.2–76.4)	53.3 (44.3–62.4)
	B	11.8 (9.8–14.3)	11.5 (9.6–13.9)	31.7 (25.3–38.1)	26.5 (21.2–31.9)	43.7 (36.3–51.1)	38.3 (31.8–44.7)
	C	7.7 (6.4–9.3)	6.5 (5.4–7.9)	27.7 (21.7–33.7)	25.9 (20.3–31.6)	35.4 (29.2–41.7)	32.6 (26.9–38.3)

Table 9 Post-hoc comparison of least-square means of nitrate (NO_3^-), ammonium (NH_4^+) and total inorganic N ($\text{NO}_3^- + \text{NH}_4^+$; N-inorg) in the soil at the depth 0–20 cm in the unit kg N ha^{-1} comparing resources on opposite sides (RO) with resources on the same side (RS) of trees and managed (TM) with unmanaged poplar trees (TU) presented as ratio or estimate with standard error (SE), z-ratio (z), and adjusted p-values (p) at each date (before sow introduction in March 2023 (1), after two batches of sows in October 2023 (2), and after three batches of sows in March 2024 (3)), and each vegetation zone (grass zone A, poplar tree zone B, and grass zone C)

Date	Zone	NO_3^- -N			NH_4^+ -N			N-inorg		
		RO/RS	SE	z	p	RO-RS	SE	z	p	p
1	A	1.09	0.11	0.85	0.397	0.17	0.18	0.95	0.340	0.27
	B	1.28	0.13	2.37	0.018	0.83	0.29	2.88	0.004	0.41
	C	1.10	0.11	0.87	0.386	-0.02	0.17	-0.10	0.922	0.25
2	A	0.72	0.12	-1.98	0.048	-6.55	2.08	-3.16	0.002	2.19
	B	1.58	0.26	2.80	0.005	3.49	1.74	2.01	0.044	2.12
	C	0.95	0.16	-0.31	0.758	5.76	2.05	2.82	0.005	2.16
3	A	0.90	0.12	-0.77	0.442	-11.87	2.92	-4.06	<0.001	2.68
	B	1.17	0.16	1.16	0.248	-0.01	1.53	-0.01	0.992	1.78
	C	1.12	0.15	0.85	0.395	7.06	1.50	4.71	<0.001	1.55
1	A	TM/TU	SE	z	p	TM-TU	SE	z	p	TM-TU
	B	0.89	0.09	-1.16	0.248	-0.38	0.18	-2.17	0.030	0.27
	C	0.78	0.08	-2.35	0.019	0.08	0.29	0.26	0.793	0.41
2	A	0.88	0.09	-1.22	0.222	0.25	0.17	1.51	0.130	0.25
	B	0.62	0.10	-2.93	0.003	-0.17	2.08	-0.08	0.935	2.19
	C	0.84	0.14	-1.04	0.296	0.45	1.74	0.26	0.795	2.12
3	A	0.98	0.16	-0.13	0.894	2.07	2.05	1.01	0.312	2.16
	B	0.78	0.11	-1.86	0.063	4.74	2.92	1.62	0.105	2.68
	C	1.02	0.14	0.18	0.860	1.85	1.53	1.21	0.227	1.93
		1.18	0.16	1.22	0.223	0.63	1.50	0.42	0.673	1.54

Table 10 Spearman's correlation indices (r_s) for nitrate (NO_3^-), ammonium (NH_4^+) and total inorganic N ($\text{NO}_3^- + \text{NH}_4^+ + \text{N-inorg}$) and the model-estimated herbaceous cover at d days before each soil sampling occasion

d	NO_3^-		NH_4^+		N-inorg	
0	−0.67	***	−0.70	***	−0.72	***
3	−0.66	***	−0.69	***	−0.71	***
7	−0.63	***	−0.66	***	−0.68	***
14	−0.54	***	−0.56	***	−0.58	***
28	−0.43	***	−0.43	***	−0.46	***
56	−0.06		−0.04		−0.06	

(initially in March 2023, after two batches of sows in October 2023, and after three batches of sows in March 2024) with indication of the significance level of p -values ($p > 0.1$ (ns), $p \leq 0.1$ (.), $p \leq 0.05$ (*), $p \leq 0.01$ (**), $p \leq 0.001$ (***)

References

- Andersen HM (2020) Pig elimination behavior—A review. *Appl Anim Behav Sci* 222:104888. <https://doi.org/10.1016/j.applanim.2019.104888>
- Andersen HML, Pedersen LJ (2011) The effect of feed trough position on choice of defecation area in farrowing pens by loose sows. *Appl Anim Behav Sci* 131:48–52. <https://doi.org/10.1016/J.APPLANIM.2011.01.005>
- Averett JM, Klips RA, Nave LE, Frey SD, Curtis PS (2004) Effects of Soil Carbon Amendment on Nitrogen Availability and Plant Growth in an Experimental Tallgrass Prairie Restoration. *Restor Ecol* 12:568–574. <https://doi.org/10.1111/J.1061-2971.2004.00284.X>
- Barrett JE, Burke IC (2000) Potential nitrogen immobilization in grassland soils across a soil organic matter gradient. *Soil Biol Biochem* 0717(00):1707–1716. [https://doi.org/10.1016/S0038-0717\(00\)00089-4](https://doi.org/10.1016/S0038-0717(00)00089-4)
- Bornett HLI, Edge HL, Edwards SA (2003) Alternatives to nose-ringing in outdoor sows. *Appl Anim Behav Sci* 159(03):267–276. [https://doi.org/10.1016/S0168-1591\(03\)00143-6](https://doi.org/10.1016/S0168-1591(03)00143-6)
- Breuning-Madsen H, Greve MH, Nørr A (2001) Danish Soil Classification and Establishment of the Danish Soil Database. *Soil Surv Horiz* 42:24. <https://doi.org/10.2136/sh2001.1.0024>
- Brooks, M.E., Kristensen, K., van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Mächler, M., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *R J* 9, 378–400. <https://doi.org/10.32614/RJ-2017-066>
- Calosi M, Gabbrielli C, Lazzeri L, Fattorini N, Cesaretti G, Burrini L, Petrillo O, Ferretti F (2024) Seasonal and Ecological Determinants of Wild Boar Rooting on Priority Protected Grasslands. *Environ Manage* 74:268–281. <https://doi.org/10.1007/S00267-024-01952-Y>
- Communication from the Commission Guidelines for State aid in the agricultural and forestry sectors and in rural areas 2022/C 485/01, 2022. Commission of the European Union.
- Dahlgren RA, Singer MJ, Huang X (1997) Oak tree and grazing impacts on soil properties and nutrients in a californian oak woodland. *Biogeochemistry* 39:45–64. <https://doi.org/10.1023/A:1005812621312>
- Davie, T., Quinn, N.W., 2019. Chapter 10: Water Quality, in: *Fundamentals of Hydrology*, p. 16.
- Dourmad JY, Guingand N, Latimier P, Sève B (1999) Nitrogen and phosphorus consumption, utilisation and losses in pig production: France. *Livest Prod Sci* 6226(99):199–211. [https://doi.org/10.1016/S0301-6226\(99\)00009-3](https://doi.org/10.1016/S0301-6226(99)00009-3)
- Ellermann, T., Bossi, R., Sørensen, M.O.B., Christensen, J., Lansø, A.S., Poulsen, M.B., 2024. Atmosfærisk deposition 2022.
- Eriksen J (2001) Implications of grazing by sows for nitrate leaching from grassland and the succeeding cereal crop. *Grass Forage Sci* 56:317–322. <https://doi.org/10.1046/j.1365-2494.2001.00279.x>
- Eriksen J, Kristensen K (2001) Nutrient excretion by outdoor pigs: a case study of distribution, utilization and potential for environmental impact. *Soil Use Manag* 17:21–29. <https://doi.org/10.1111/j.1475-2743.2001.tb00004.x>
- Eriksen J, Petersen SO, Sommer SG (2002) The fate of nitrogen in outdoor pig production. *Agronomie* 22:863–867. <https://doi.org/10.1051/agro:2002045>
- Eriksen J, Studnitz M, Strudsholm K, Kongsted AGG, Hermansen JEE (2006) Effect of nose ringing and stocking rate of pregnant and lactating outdoor sows on exploratory behaviour, grass cover and nutrient loss potential. *Livest Sci* 104:91–102. <https://doi.org/10.1016/j.livsci.2006.03.008>
- Fortier J, Gagnon D, Truax B, Lambert F (2010) Nutrient accumulation and carbon sequestration in 6-year-old hybrid poplars in multiclinal agricultural riparian buffer strips. *Agric Ecosyst Environ* 137:276–287. <https://doi.org/10.1016/j.agee.2010.02.013>
- Fortier J, Truax B, Gagnon D, Lambert F (2013) Root biomass and soil carbon distribution in hybrid poplar riparian buffers, herbaceous riparian buffers and natural riparian woodlots on farmland. *Springerplus* 2:1–19. <https://doi.org/10.1186/2193-1801-2-539>
- Fortier J, Truax B, Gagnon D, Lambert F (2015) Biomass carbon, nitrogen and phosphorus stocks in hybrid poplar buffers, herbaceous buffers and natural woodlots in the riparian zone on agricultural land. *J Environ Manage* 154:333–345. <https://doi.org/10.1016/j.jenvman.2015.02.039>

- Gao S, Hendratna A, Thao T, Culumber CM, Poret-Peterson AT, Zuber CAT, Holtz BA (2024) Influence of wood-chip size and nitrogen fertilization on carbon dioxide and nitrous oxide emissions from soils amended with orchard biomass. *Soil Sci Soc Am J* 88:803–815. <https://doi.org/10.1002/SAJ2.20650>
- Hadrović, S., Jovanović, F., Braunović, S., Mitrović, T.Ć., Rakonjac, L., Jandri, M., Hadrovi, D., 2022. Biomass Carbon and Nitrogen Content of Softwood Broadleaves in Southwestern Serbia. *HortScience* 57, 684–685. <https://doi.org/10.21273/HORTSCI16457-21>
- Hartig, F., 2022. Dharma: Residual diagnostics for hierarchical (multi-level / mixed) regression models [R package version 0.4.6].
- Horsted K, Kongsted AGG, Jørgensen U, Sørensen J (2012) Combined production of free-range pigs and energy crops—animal behaviour and crop damages. *Livest Sci* 150:200–208. <https://doi.org/10.1016/j.livsci.2012.09.006>
- Jakobsen M, Preda T, Kongsted AG, Hermansen JE (2015) Increased foraging in outdoor organic pig production—modeling environmental consequences. *Foods* 4:622–644. <https://doi.org/10.3390/foods4040622>
- Jakobsen M, Hermansen JE, Andersen HM-L, Jørgensen U, Labouriau R, Rasmussen J, Kongsted AG (2018) Elimination behavior and soil mineral nitrogen load in an organic system with lactating sows – comparing pasture-based systems with and without access to poplar (*Populus sp.*) trees. *Agroecol Sustain Food Syst* 43:639–661. <https://doi.org/10.1080/21683565.2018.1541039>
- Jakobsen, M., 2018. Integrating foraging and agroforestry into organic pig production - environmental and animal benefits (PhD Thesis). Aarhus University.
- Jan P, Calabrese C, Lips M (2017) Determinants of nitrogen surplus at farm level in Swiss agriculture. *Nutr Cycl Agroecosyst* 109:133–148. <https://doi.org/10.1007/S10705-017-9871-9>
- Jensen MB, Pedersen LJ (2007) The value assigned to six different rooting materials by growing pigs. *Appl Anim Behav Sci* 108:31–44. <https://doi.org/10.1016/J.APPLA NIM.2006.10.014>
- Jørgensen U, Thuesen J, Eriksen J, Horsted K, Hermansen JE, Kristensen K, Kongsted AG (2018) Nitrogen distribution as affected by stocking density in a combined production system of energy crops and free-range pigs. *Agrofor Syst* 92:987–999. <https://doi.org/10.1007/s10457-018-0200-3>
- Kongsted, A.G., Pedersen, B.F., Kristensen, I.S., Kristensen, T., Eriksen, J., 2019. Miljøpåvirkning fra udendørs hold af grise – Del 1 (No. 2019–0023177). DCA - Nationalt Center for Fødevarer og Jordbrug.
- Kongsted, A.G., Kristensen, I.S., Pedersen, B.F., Eriksen, J., Kristensen, T., 2020. Miljøpåvirkning fra udendørs hold af grise – Del 2. DCA - Nationalt Center for Fødevarer og Jordbrug.
- Landbrugsstyrelsen, 2022. Vejledning om tilskud til vådområde- og lavbundsprojekter 2022.
- Landbrugsstyrelsen, 2025. Vejledning om økologisk jordbrugsproduktion. Ministeriet for Fødevarer, Landbrug og Fiskeri, København.
- Ma L, Xu X, Zhang C, Lv Y, Liu G, Zhang Q, Feng J, Wang R (2021) Strong non-growing season N uptake by deciduous trees in a temperate forest: A ¹⁵N isotopic experiment. *J Ecol* 109:3752–3766. <https://doi.org/10.1111/1365-2745.13754>
- Manevski K, Jakobsen M, Kongsted AG, Georgiadis P, Labouriau R, Hermansen JE, Jørgensen U (2019) Effect of poplar trees on nitrogen and water balance in outdoor pig production – A case study in Denmark. *Sci Total Environ* 646:1448–1458. <https://doi.org/10.1016/j.scitotenv.2018.07.376>
- Manevski K, Steinfeldt S, Hellwing ALF, Andersen HML, Jørgensen U (2024) Nitrate leaching and nitrogen balances for integrated willow-poultry organic systems in Denmark. *Agric Syst* 221:104149. <https://doi.org/10.1016/J.AGSY.2024.104149>
- Meier HEM, Eilola K, Almroth-Rosell E, Schimanke S, Kniebusch M, Höglund A, Pemberton P, Liu Y, Väli G, Saraiva S (2019) Disentangling the impact of nutrient load and climate changes on Baltic Sea hypoxia and eutrophication since 1850. *Clim Dyn* 53:1145–1166. <https://doi.org/10.1007/s00382-018-4296-y>
- Pecenka R, Lenz H, Hering T (2020) Options for optimizing the drying process and reducing dry matter losses in whole-tree storage of poplar from short-rotation coppices in Germany. *Forests* 2020(11):374. <https://doi.org/10.3390/F11040374>
- R Core Team, 2024. R: A Language and Environment for Statistical Computing.
- Rubæk, G., Sørensen, P., 2011. Jordanalyser - kvalitet og anvendelse. Rep. No. 002. Aarhus University, DCA - Nationalt Center for Fødevarer og Jordbrug. Blichers Allé 20, postboks 50, 8830 Tjele, Denmark.
- Salomon E, Åkerhielm H, Lindahl C, Lindgren K (2007) Outdoor pig fattening at two Swedish organic farms-Spatial and temporal load of nutrients and potential environmental impact. *Agric Ecosyst Environ* 121:407–418. <https://doi.org/10.1016/j.agee.2006.11.017>
- Schild SA, Rangstrup-Christensen L, Bonde M, Pedersen LJ (2018) The use of a shaded area during farrowing and lactation in sows kept outdoors. *Appl Anim Behav Sci* 209:22–29. <https://doi.org/10.1016/j.applanim.2018.08.019>
- Schullehner J, Hansen B, Thygesen M, Pedersen CB, Sigsgaard T (2018) Nitrate in drinking water and colorectal cancer risk: A nationwide population-based cohort study. *Int J Cancer* 143:73–79. <https://doi.org/10.1002/IJC.31306>
- Seneviratne G, Van Holm LHJ, Kulasoorya SA (1997) Quality of different mulch materials and their decomposition and N release under low moisture regimes. *Biol Fertil Soils* 26:136–140. <https://doi.org/10.1007/S003740050356/METRCS>
- Sønsteby A, Nes A, Måge F (2004) Effects of bark mulch and npk fertilizer on yield, leaf nutrient status and soil mineral nitrogen during three years of strawberry production. *Acta Agriculturae Scandinavica, Section B - Soil & Plant Science* 54:128–134. <https://doi.org/10.1080/09064710410030276>
- Stolba A, Wood-Gush DGM (1989) The behaviour of pigs in a semi-natural environment. *Anim Prod* 48:419–425. <https://doi.org/10.1017/S0003356100040411>

- Tabeling R, Schwier S, Kamphues J (2003) Effects of different feeding and housing conditions on dry matter content and consistency of faeces in sows. *J Anim Physiol Anim Nutr (Berl)* 87:116–121. <https://doi.org/10.1046/J.1439-0396.2003.00423.X>
- Tahboub MB, Lindemann WC, Murray L (2007) Nutrient availability in soil amended with pecan wood chips. *HortScience* 42:339–343. <https://doi.org/10.21273/HORTSCI.42.2.339>
- Tang Q, Wang J, Cao M, Chen Z, Tu X, Elrys AS, Jing H, Wang X, Cai Z, Müller C, Daniell TJ, Yan X, Cheng Y (2024) Awakening soil microbial utilization of nitrate by carbon regulation to lower nitrogen pollution. *Agric Ecosyst Environ* 362:108848. <https://doi.org/10.1016/J.AGEE.2023.108848>
- Thevathasan NV, Gordon AM (1997) Poplar leaf biomass distribution and nitrogen dynamics in a poplar-barley intercropped system in southern Ontario, Canada. *Agrofor Syst* 37:79–90. <https://doi.org/10.1023/A:1005853811781>
- Williams JR, Chambers BJ, Hartley AR, Ellis S, Guise HJ (2000) Nitrogen losses from outdoor pig farming systems. *Soil Use Manag* 16:237–243. <https://doi.org/10.1111/j.1475-2743.2000.tb00202.x>
- Yansheng C, Fengliang Z, Zhongyi Z, Tongbin Z, Huayun X (2020) Biotic and abiotic nitrogen immobilization in soil incorporated with crop residue. *Soil Tillage Res.* <https://doi.org/10.1016/j.still.2020.104664>

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